Guided Notes: Module 7 Explaining Application Services

Module Objectives

- Understand the importance of **time synchronization** and the role of **Network Time Protocol (NTP)**.
- Explain key application services: **web services, file/print services, and database services**.
- Describe **email and voice services**.
- Understand **high availability** concepts including **redundancy** and **load balancing** for service provisioning.

Lesson 7.1: Application Security and Time Synchronization

Transport Layer Security (TLS)

- TLS is a protocol that **encrypts** data to protect it from interception.
- It provides **secure communication** between clients and servers.

TLS Handshake Process

- Steps in establishing a secure TLS connection:
    1. **Client Hello:** Client initiates communication, proposing security options.
    2. **Server Hello:** Server responds with chosen options.
    3. **Client Key Exchange:** Client sends key material.
    4. **Session Key Established:** Both parties establish a shared secret key.
    5. **Server Finished:** Server confirms handshake completion.

Network Time Protocol (NTP)

- NTP is used to **synchronize clocks across devices** on a network to Coordinated Universal Time (UTC).
- Accurate time synchronization is crucial for logging, security protocols, and network troubleshooting.

Precision Time Protocol (PTP)

- PTP is a more precise time synchronization protocol often used in industrial and telecommunications networks.
- The **Grandmaster Clock** acts as the main time source in the PTP domain.

Lesson 7.2: Web, File/Print, and Database Services

Hypertext Transfer Protocol (HTTP)

- Protocol for transferring web pages.
- Process overview:
    1. User enters a URL.
    2. Browser translates URL into an HTTP request.
    3. The server interprets the request and retrieves data.
    4. The server sends an HTTP response with status code and data.
    5. Browser renders and displays the webpage.

HTTPS (HTTP Secure)

- HTTP over TLS/SSL for secure web communication on port 443.
- Encrypts data to protect against eavesdropping and tampering.

File Transfer Protocols

- **FTP:** Non-secure file transfer protocol.
- **SFTP:** Secure version of FTP using SSH for encryption.

Server Message Block (SMB) Protocol

- Used for sharing files, printers, and other resources in Windows networks.
- Clients request access; servers grant or deny access to SMB shared resources.

Network Attached Storage (NAS)

- Dedicated file storage device connected to a network.
- Provides shared storage accessible by multiple clients.

Database Services

- **Relational Database Management Systems (RDBMS):**
  - Use structured tables with predefined schemas.
  - Support SQL queries.
- **NoSQL Databases:**
  - Use flexible schema or schema-less design.
  - Useful for large-scale, distributed data storage.

RDBMS vs NoSQL

- RDBMS is structured, transactional, and uses SQL.
- NoSQL is flexible, scalable, and suited for big data or real-time web apps.

Lesson 7.3: Email and Voice Services

Simple Mail Transfer Protocol (SMTP)

- Protocol used to **send emails** between servers.

Internet Message Access Protocol (IMAP)

- Protocol used to **access and manage emails remotely** on the mail server.
- Allows synchronization across multiple devices.

SMTP and IMAP Interaction Diagram

- Sender’s email server uses SMTP to send mail to receiver’s email server.
- Receiver accesses mail using IMAP.

Voice over IP (VoIP) Protocols

- Protocols that enable voice and video communication over IP networks.
- Examples include SIP, RTP, and others.

Trivia Activity

- Protocol to remotely manage emails: **IMAP**
- Protocol to send emails between servers: **SMTP**

Lesson 7.4: Disaster Recovery and High Availability

Disaster Recovery Concepts

- Planning to restore IT systems after disruptions.
- Includes strategies for backup, failover, and recovery sites.

Disaster Recovery Metrics

- Measures like Recovery Time Objective (RTO) and Recovery Point Objective (RPO) define acceptable downtime and data loss.

Disaster Recovery Sites

- Types include:
    - **Hot Site:** Fully operational, immediate failover.
    - **Warm Site:** Partially ready, some setup needed.
    - **Cold Site:** Basic infrastructure, longer setup time.

Fault Tolerance and Redundancy

- Use of duplicate hardware and software to ensure continuous operation.
- Redundancy avoids single points of failure.

Key Performance Indicators (KPIs)

- Metrics to assess system reliability, uptime, and performance.

Load Balancers

- Distribute network or application traffic across multiple servers.
- Improve responsiveness and availability.

High Availability Clusters

- Groups of servers working together to provide continuous service.
- If one server fails, others take over.

First Hop Redundancy Protocols

- Ensure continuous gateway availability for devices by providing backup routers.

Hot, Warm, or Cold Site Activity

- For a financial services company requiring minimal downtime and high transaction volume, a **Hot Site** is best.

Summary: Explaining Application Services

Main Points

- Implement **HTTPS everywhere** using TLS on web servers.
- Use secure file transfer protocols like **SFTP** or **FTPS**.
- Limit unencrypted services such as SMB to trusted internal networks.
- Develop **Disaster Recovery (DR)** and **High Availability (HA)** plans with dedicated resources to ensure business continuity.

Additional Notes and Tips

- Always ensure time synchronization (NTP/PTP) for network reliability and security.
- Secure application services protect data integrity and privacy.
- Understand differences between database types to choose the right one for your needs.
- Disaster recovery planning is essential for minimizing downtime and data loss.
- Use redundancy and load balancing to maintain high availability of critical services.

Vocabulary List: Explaining Application Services:

Application Services
Services that provide networked applications such as web, email, file sharing, and database access.

Transport Layer Security (TLS)
A protocol that encrypts data to secure communications over a network.

TLS Handshake
The process of establishing a secure TLS connection between a client and server.

Network Time Protocol (NTP)
A protocol used to synchronize clocks across devices to Coordinated Universal Time (UTC).

Precision Time Protocol (PTP)
A high-precision time synchronization protocol used in specialized networks.

Grandmaster Clock
The primary time source in a PTP domain.

Hypertext Transfer Protocol (HTTP)
The protocol used for transferring web pages over the internet.

HTTPS (HTTP Secure)
The secure version of HTTP that uses TLS to encrypt data.

File Transfer Protocol (FTP)
A protocol used for transferring files, typically unencrypted.

Secure File Transfer Protocol (SFTP)
A secure version of FTP that uses SSH encryption.

Server Message Block (SMB)
A network protocol for sharing files, printers, and other resources in a Windows environment.

Network Attached Storage (NAS)
A dedicated file storage device connected to a network providing shared access.

Relational Database Management System (RDBMS)
A database system that stores data in tables with predefined schemas and supports SQL.

NoSQL Database
A database system that uses flexible, schema-less data storage, often for large-scale or unstructured data.

Simple Mail Transfer Protocol (SMTP)
The protocol used to send emails between mail servers.

Internet Message Access Protocol (IMAP)
A protocol used to access and manage emails stored on a mail server remotely.

Voice over IP (VoIP)
Technology that allows voice and video communications to be transmitted over IP networks.

Disaster Recovery (DR)
Plans and processes to restore IT systems after disruptive events.

Recovery Time Objective (RTO)
The maximum acceptable amount of time to restore a system after a failure.

Recovery Point Objective (RPO)
The maximum acceptable amount of data loss measured in time.

Disaster Recovery Site
A location equipped to restore IT operations. Types include hot, warm, and cold sites.

Fault Tolerance
The ability of a system to continue operating despite failures.

Redundancy
Duplication of critical components to increase reliability and availability.

Key Performance Indicators (KPIs)
Metrics used to evaluate the performance and reliability of systems.

Load Balancer
A device or software that distributes network or application traffic across multiple servers.

High Availability Cluster
A group of servers working together to provide continuous service despite failures.

First Hop Redundancy Protocol
Protocols that provide automatic failover between routers to ensure gateway availability.

Acronym List: Explaining Application Services:

TLS — Transport Layer Security
NTP — Network Time Protocol
PTP — Precision Time Protocol
HTTP — Hypertext Transfer Protocol
HTTPS — Hypertext Transfer Protocol Secure
FTP — File Transfer Protocol
SFTP — Secure File Transfer Protocol
SMB — Server Message Block
NAS — Network Attached Storage
RDBMS — Relational Database Management System
NoSQL — Not Only SQL (Non-relational Database)
SMTP — Simple Mail Transfer Protocol
IMAP — Internet Message Access Protocol
VoIP — Voice over Internet Protocol
DR — Disaster Recovery
RTO — Recovery Time Objective
RPO — Recovery Point Objective
KPI — Key Performance Indicator